

Instructions:

测试说明:

- The total time cap will be 15 minutes. Please note how much time you took.
- 测试总时长为 15 分钟，若提前完成请标注使用时长。
- The first 10 questions are fundamentals for the class.
- 前十道为相对基础的题目。
- The last two are hard problems that are not intended for you to answer accurately. They are a reference for me to check if I would need to raise the class difficulty.
- 后两道为相对难的题目，未能作答是非常正常的，主要是帮助我来确认是否需要提高课程难度。



1. Solve for x 解方程: $3(x - 4) + 5 = 2x + 7$
- a) 7
 - b) 14
 - c) 12
 - d) 16
2. What is the sum of the two roots of the equation 此方程的两个根之和为多少: $x^2 - 5x + 6 = 0$
- a) -6
 - b) 6
 - c) 5
 - d) -5
3. What is the biggest integer n satisfying the inequality 最大的满足此不等式的整数 n 未多少: $4n - 7 < 18$
- a) 7
 - b) 5
 - c) 8
 - d) 6
4. Calculate 计算: $\frac{1}{2} + \frac{1}{3} + \frac{1}{6}$
- a) $\frac{7}{6}$
 - b) $\frac{1}{11}$
 - c) $\frac{5}{6}$
 - d) 1
5. What is the measure (in degrees) of any angle of a regular hexagon 一个正六边形任意一个内角的度数为多少?
- a) 150°

b) 120°

c) 108°

d) 135°

6. If a circle has an area of 16π , what is its perimeter 若一个圆面积为 16π , 其的周长为多少?

a) 32π

b) 16π

c) 8π

d) 4π

7. What is the least common multiple of 12,15, and 20 这三个数字的最小公倍数为多少?

a) 30

b) 240

c) 120

d) 60

8. If you toss two regular dice at once, what is the probability the sum of the numbers rolled is 7 若你在同时投掷两个正常骰子, 正面朝上的数字之和为 7 的概率是多少?

i. 提示: 尝试枚举在多少种朝上数字组合下它们的和为 7

b) $\frac{1}{7}$

c) $\frac{1}{6}$

d) $\frac{1}{12}$

e) $\frac{1}{36}$

9. What is the distance between points (1, 2) and (3, 6) in the Cartesian Plane 平面直角坐标系中, 点 (1, 2) 和 (3, 6) 的距离为多少?

- a) 5
- b) $2\sqrt{5}$
- c) $\sqrt{6}$
- d) $2\sqrt{3}$

10. What is the 10th number in the following arithmetic sequence:
2,5,8,11...这个等差数列中第十项为多少。

- a) 26
- b) 27
- c) 28
- d) 29

11. (Hard 难) Alice and Bob's ages currently fulfill the ratio 3:4. After 5 years, their age ratio will become 4:5. How old is Alice 小红和小明今年的岁数之比为 3: 4。5 年后，他们的岁数之比将会变为 4: 5。今年小红多少岁？

- a) 3
- b) 15
- c) 27
- d) 12

12. (Hard 难) What is the remainder when dividing 10^{2026} by 9 请问 10^{2026} 除以 9 的余数为多少？

- a) 2
- b) 1
- c) 8
- d) 7